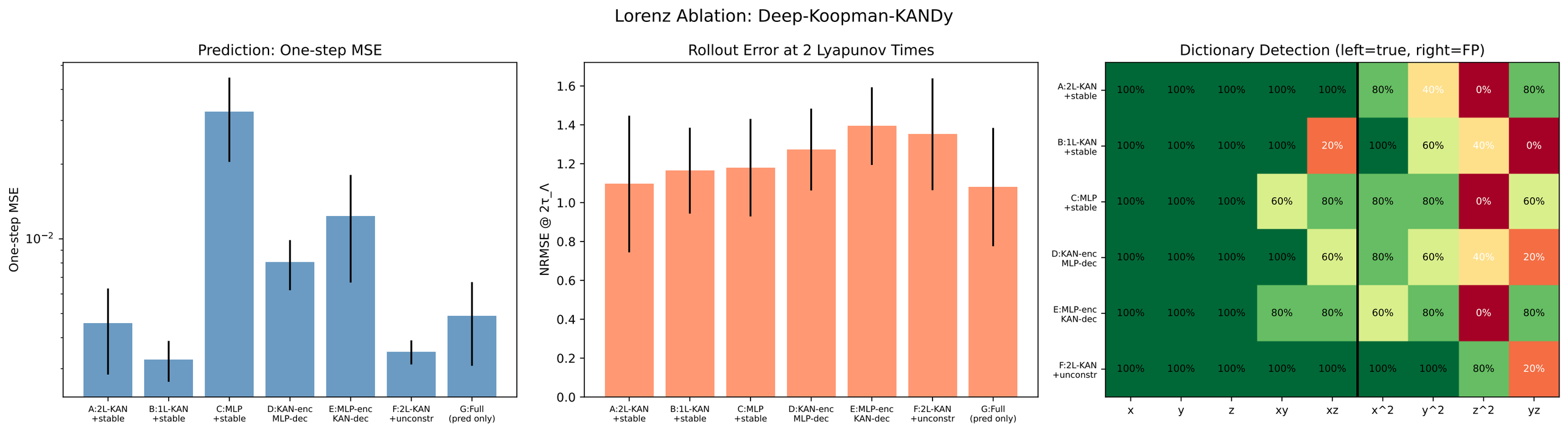
Lorenz Ablation: Deep-Koopman-KANDy
Prediction: One-step MSE
One-step MSE
10^{-2}
A:2L-KAN +stable
B:1L-KAN +stable
C:MLP +stable
D:KAN-enc MLP-dec
E:MLP-enc KAN-dec
F:2L-KAN +unconstr
G:Full (pred only)
Rollout Error at 2 Lyapunov Times
NRMSE @ 2τ_Λ
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Dictionary Detection (left=true, right=FP)
A:2L-KAN +stable
100% 100% 100% 100% 100% 80% 40% 0% 80%
B:1L-KAN +stable
100% 100% 100% 100% 20% 100% 60% 40% 0%
C:MLP +stable
100% 100% 100% 60% 80% 80% 80% 0% 60%
D:KAN-enc MLP-dec
100% 100% 100% 100% 60% 80% 60% 40% 20%
E:MLP-enc KAN-dec
100% 100% 100% 80% 80% 60% 80% 0% 80%
F:2L-KAN +unconstr
100% 100% 100% 100% 100% 100% 100% 80% 20%
x
y
z
xy
xz
x^2
y^2
z^2
yz